

RESUME

VINAY M. MADGI

Hubballi, Karnataka, India

+91 7975200870 | vinaymadgi28@gmail.com

[LinkedIn](#) | [GitHub](#)



Career Objective

Computer Science undergraduate at **KLE Technological University** with interests in **AI/ML, Generative AI, Data Engineering and Backend Development**. Seeking opportunities in **AI/ML, Data Engineering, SDE or Backend Development** to build scalable, intelligent software systems.

Education

B.E. in Computer Science & Engineering — KLE Technological University, Hubballi
2023–2027 | **CGPA: 8.36** (Semesters 1–6) | No Backlogs

PUC — Pre-University Board, Bengaluru | 2023 | **70%**

Class X — CBSE, New Delhi | 2021 | **68%**

Technical Skills

- **Languages:** Python, Java, C++, SQL
- **Web/Database:** Django, MongoDB, MySQL, REST APIs
- **AI/ML:** PyTorch, Transformers, LangChain, FAISS, RAG, Meta-Learning, Vision–Language Models
- **DevOps/Tools:** Docker, Kubernetes, Jenkins, Git, GitHub, Postman, Prometheus, Grafana
- **Core CS:** DSA, OOP, DBMS, Operating Systems

Projects

7th Semester — Ongoing

Beyond Byte-Level BERT: Protocol-Aware NIDS using DistilBERT | REU

- Developing a protocol-aware **Network Intrusion Detection System** using structured TCP flags, service ports, and bidirectional flow representations.
- Fine-tuning **DistilBERT** for DoS, web, and application-layer attack detection with emphasis on zero-day generalization and low-latency inference.

EnergyCast: Electricity Price Forecasting System

- Developing an ML-based forecasting system for **IEX Day-Ahead Market** prices using demand, renewable generation, weather, and temporal features.
- Engineering lag and rolling features with chronological model evaluation and a **Streamlit dashboard** for forecast visualization.

6th Semester — Completed

Citation-Grounded Academic Report Generation using GraphRAG & NLI

- Built a citation-grounded report generation pipeline using **OCR, BGE embeddings, FAISS, GraphRAG, and LLMs**.
- Integrated hierarchical planning and **NLI-based fact verification with reflection** to improve factual consistency and citation reliability.

5th Semester — Completed

Parameter-Efficient Vision–Language Model Adaptation for Fashion Retrieval

- Built an image–text retrieval framework using frozen **EVA02-CLIP/CLIP** encoders and a lightweight projection head.

- Evaluated **44,441 samples** with **<2.4M trainable parameters** using a hybrid regression and contrastive objective.

Amortized Hyperparameter Optimization for Few-Shot Classification

- Built a **Static Meta-Learning** framework for predicting hyperparameters without repeated per-task search.
- Evaluated **800 tasks across 5 datasets and 4 learning regimes**, achieving **62.964 ± 0.019% held-out accuracy**.

Publications

Journal Manuscripts

- **Parameter-Efficient Adaptation of Vision–Language Models for Domain-Specific Multimedia Retrieval in Fashion** — Published in *Multimedia Tools and Applications*, Springer. (Volume 85, article number 793(2026), <https://doi.org/10.1007/s11042-026-21961-9>)
- **Static Meta-Learning for One-Shot Hyperparameter Prediction in Few-Shot Image Classification** — Communicated to *Neural Computing and Applications*, Springer; decision pending.

Conference Publications

- **Edge-Native IoT Smart Surveillance System with Multi-Factor Threat Reasoning** — Presented at **CICBA 2026**; accepted for publication in **Springer CCIS**.
- **Analysis of Factors Causing Infant Mortality in Karnataka and Recommendations for Prevention** — Published in **IEEE Xplore**, **INSPIRE 2025**. DOI: **10.1109/INSPIRE67328.2025.11300582**

Certifications

- **Jenkins for DevOps: Jenkins Configuration for DevOps** — Infosys Springboard, 2025
- **DevOps Mindset: Understanding DevOps Tools** — Infosys Springboard, 2025

Achievements

- Presented research at **CICBA 2026** and **ICACIT 2026**, with accepted Springer conference publications.
- Published research in **IEEE Xplore** through the **INSPIRE 2025** conference.
- Engaged in a **Research Experience for Undergraduates (REU) project** focused on AI-driven Network Intrusion Detection.

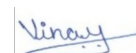
Areas of Interest

Artificial Intelligence & Machine Learning | Generative AI & LLMs | Data Engineering & Analytics | Backend Engineering

Address for Communication

H. No. 111, First Floor, 3rd Cross, Laxmi Nagar, Gokul Road, Hubballi 580030, Karnataka, India

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Signature